Histology of the Testis and Sperm Release Pathway

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Textbook Link

Histology: A Text and Atlas, 9th ed.–Chapter 22: Male Reproductive System

Learning Objectives

By exam time, you should be able to do the following:

1. Discuss the structural and functional anatomy of the testis and the organs involved in sperm maturation and release, including how they change over different life stages and the hormonal regulation of these changes.

2. Describe and identify the cells and tissues of the testis and organs involved with sperm maturation and release by histology, including how those structures change over different life stages and via hormonal changes.

3. Discuss the process of spermatogenesis from germ cell through mature sperm, including the different sites in the reproductive tract where the steps of maturation occur.

4. Predict how different pathologies of the testis and organs involved in sperm maturation and release might alter the histological or physiologic features of the component cells/tissues.

Study Questions

1. Describe histologic features of the testes in post-pubertal individuals.

2. Discuss the histology and significance of the blood–testis barrier.

3. Compare and contrast histologic features of the genital ducts involved in sperm transport and post-testicular sperm maturation.

4. Compare and contrast histologic features of seminal vesicles and the prostate gland.

Biolucida Slide Box Hyperlink

Folder: Histology of the Testis and Sperm Release Pathway

Slides Used in This Lab	Extra Practice Slides in the Biolucida Slide Box Folder
Testis VS 1, Human, H&E	**Testis Extra Practice VS 1**, Human, H&E
Testis VS 2, Human, H&E	**Testis Extra Practice VS 2**, Human, H&E
Epididymis VS 1, unknown species, H&E	**Testis Extra Practice VS 3**, unknown species, H&E
Epididymis VS 2, Hamster, H&E	**Epididymis Extra Practice VS 1**, Monkey, H&E
Spermatic Cord VS 1, unknown species, H&E	**Ductus Deferens Extra Practice VS 1**, Human, H&E
Seminal Vesicle VS 1, Monkey, H&E	**Seminal Vesicle Extra Practice VS 1**, Human, H&E
Prostate VS 1, Human, H&E	**Prostate Extra Practice VS 1**, Human, H&E
Penis VS 1, unknown species, H&E	

Instructions: This handout focuses on the questions posed in the linked Biolucida slides. It is recommended that you review the pre-lab PowerPoint slides and/or their associated narrative slide notes before beginning this lab exercise.

During the lab exercise, you should view the questions in the annotated slides and work with your group to discuss the answers. After this session, an answer key will be released on Canvas.

In addition to the slides used for questions in this lab, several additional slides have been populated in the associated slide folder for additional practice in histological identification.

Accessibility Note for Students with Color Vision Deficiency: While the questions often refer to a "red polygon," "green arrow," etc., each question also has an identifying encircled number on the slide.

A. Testis VS 1

Slide link

1. What is the single cell layer indicated by the solid black arrows? What type of epithelium is it composed of? What is the function of this structure?
   This is the tunica vaginalis; specifically, its visceral layer. The parietal layer of the tunica vaginalis is the deepest layer of the scrotal wall. The lining of the tunica vaginalis is a mesothelium: a simple squamous (to cuboidal) epithelium producing a trace amount of fluid, allowing structures to move in a well lubricated fashion within a body cavity. (There are parallels to the pericardium, pleura, and peritoneum.)

2. What is the structure indicated by the dashed black lines? What type of tissue is it composed of? What is its relationship to the structure indicated with the double black arrows?
   This is the tunica albuginea, a thick layer of dense, irregular connective tissue. Immediately deep to it is the less conspicuous tunica vasculosa, with profiles of vascular structures. The division between these layers is sometimes ill-defined. The double arrows point to a fibrous septum dividing the testis into one of its roughly 250–300 lobules.

3. What is the structure in the green circle?
   This is a seminiferous tubule. The germ cells are located here and undergo meiosis and spermiogenesis.

4. What portion of the organ is indicated by the red rectangle? What is occurring in this region?
   This is the mediastinum testis; basically, the hilum of the testis. Vascular and lymphatic structures, and nerves enter and exit here. The spermatozoa are drained into the rete testis, which will subsequently drain into the efferent ductules to reach the head of the epididymis.

5. What is the structure marked "A"? What about "B"? How might you distinguish these similar-appearing structures?
   "A" indicates the rete testis, lined by a cuboidal epithelium. (This epithelium may be variable from squamous to columnar; tracing along the wall is likely to reveal at least some taller cells.) "B" indicates a vein, consistently lined with a simple squamous endothelium. The difference in the lining helps distinguish these structures; if their contents were visible, that might help too—the rete carries spermatozoa, while the veins carry blood.

6. What are the tubules indicated by the blue rectangle? What cells form their epithelium?
   These are straight tubules, which link the ends of the U-shaped seminiferous tubules to the rete testis. They are lined by Sertoli cells only, without developing germ cells.

B. Testis VS 2

Slide link

1. What type of cell is indicated is the structure shown within the black rectangle? How do connections between cells of this type subdivide the tubular lumen? What functions do these cells serve?
   This is a Sertoli cell. These columnar cells support the developing sperm as they undergo meiosis and spermiogenesis. The blood–testis barrier divides the tubular lumen of the seminiferous tubule into two compartments: the basal compartment, which contains diploid spermatogonia; and the luminal compartment, which contains cells undergoing meiosis and chromosomal rearrangement (which are therefore expressing novel proteins that may be interpreted as "foreign" by the immune system). The blood–testis barrier thereby protects the developing spermatozoa from immune attack.

   Some other important functions include:
   - Physically supporting developing spermatocytes and spermatids until they complete spermiation
   - Concentrating testosterone and other androgens in the tubular lumen
   - Phagocytosing faulty sperm cells and residual bodies (released vesicles of cytoplasm during the maturation phase of spermiation)

2. Compare and contrast the cells indicated by "A," "B," "C," and "D" on the following criteria: cell name and characteristic morphology, ploidy (haploid/diploid), and motility.

Label	*Cell name/morphology*	*Ploidy*	*Motility*
A	*Spermatogonium (type B): clumped chromatin at periphery*	*Diploid*	*Immotile*
B	*Primary spermatocyte: condensed chromosomes (about to enter meiosis I)*	*Diploid*	*Immotile*
C	*Early spermatid: decondensed chromatin but still relatively large nucleus*	*Haploid*	*Immotile*
D	*Late spermatid to spermatozoon: markedly shrunken, dark nucleus; well-formed flagellum*	*Haploid*	*Immotile (motility not gained until epididymis)*

3. What cells are indicated by the blue arrow? What do these cells produce? What stimulates their activity?
   These are interstitial (Leydig) cells, with copious eosinophilic cytoplasm. They are located between seminiferous tubules and produce testosterone under luteinizing hormone stimulation. Leydig cells also produce oxytocin, which stimulates myoid cells in the walls of the seminiferous tubules to propel the spermatozoa through peristalsis.

4. What cells, with elongated nuclei, are indicated by the red arrow? What is their function?
   These are myoid cells, which are contractile. They propel spermatozoa from the seminiferous tubules toward the straight tubules.

C. Epididymis VS 1

Slide link

1. What are the tubules indicated with the black rectangle? What structures do they connect?
   These are efferent ductules, identifiable by their scalloped or saw-toothed luminal cross section. Efferent ductules conduct spermatozoa from the rete testis to the head of the epididymis. They are lined with a pseudostratified columnar epithelium with some cilia and are surrounded by smooth muscle.

2. What are the tubules indicated with the red circle? How would you describe the epithelial lining?
   This is the epididymis proper, lined with pseudostratified stereociliated columnar epithelium. Stereocilia are long, branched microvilli ("–cilia" is a misnomer) that are involved in fluid absorption. These cells absorb testicular fluid, concentrating the spermatozoa.

D. Epididymis VS 2

Slide link

1. Unlike the previous virtual slide, this section of the epididymis does not contain efferent ductules. Given that, this section was likely taken from which region(s) of the epididymis?
   Likely the body or tail. The head of the epididymis accepts spermatozoa traveling through the efferent ductules. The body and tail do not have efferent ductules. During the transition from head to body to tail, the smooth muscle coat of the epididymis thickens.

2. What are the structures indicated by the black arrow? How do they change during their time in this organ?
   These are spermatozoa. The epididymis accumulates spermatozoa through the efferent ductules and stores them until emission and ejaculation. In the epididymis, spermatozoa gain motility and zona pellucida-binding proteins (to facilitate subsequent fertilization), but are also prevented from wasting their energy stores through the action of immobilin, a large viscoelastic glycoprotein that prevents their movement. Similarly, sperm decapacitating factors prevent premature activation of the acrosomal reaction. Abnormal spermatozoa are also phagocytosed here in the epididymis.

E. Spermatic Cord VS 1

Slide link

1. What is the structure within the black circle? What type of epithelium, indicated by the black arrow, lines its internal surface?
   This is the ductus (vas) deferens. It is lined with a pseudostratified, partially stereociliated columnar epithelium.

2. What are the layers indicated "A," "B," and "C"? What are their relative orientations?
   These are layers of smooth muscle surrounding the ductus deferens. "A" is the innermost longitudinal layer; "B" is the internal circular layer; and "C" is the outer longitudinal layer.

3. What are the structures indicated with the red rectangle? Given the identity of the tissue in the previous question, what type of fibers do they predominantly carry? What portion of the sexual response cycle do they perform?
   These are nerves, which are predominantly composed of postganglionic sympathetic fibers. This autonomic pathway previously synapsed in the inferior hypogastric plexus. These fibers cause emission, the movement of spermatozoa from the epididymis to the ejaculatory duct through peristaltic movements of the ductus deferens.

4. What type of tissue is shown in the blue circle? What is the identity of this tissue, grossly? From what structure of the abdominal wall did it derive? What is its innervation?
   This is skeletal muscle. Grossly, this is the cremaster muscle, which is derived from the muscular fibers of the internal oblique as the testis traverses the inguinal canal. The cremaster muscle receives somatic (skeletal muscle) motor innervation from the genital branch of the genitofemoral nerve.

5. What are the structures indicated by with the red and blue dashed arrows?
   The red dashed arrow is the testicular artery, which branches directly from the aorta slightly inferior to the renal arteries.
   The blue dash arrows are portions of the pampiniform plexus of veins, which will coalesce to form the testicular veins. The left testicular vein drains into the left renal vein; the right testicular vein drains into the inferior vena cava.

F. Seminal Vesicle VS 1

Slide link

1. What is this organ, indicated with the black rectangle? What is its function?
   This is the seminal vesicle, organized into lobules. It generates approximately 65–75% of the ejaculate volume. It secretes fructose, which provides a source of energy to the spermatozoa; anti-agglutination factors to prevent sperm sticking together within the genital ducts system; pro-coagulative factors (like seminogelins), which cause seminal fluid to thicken in viscosity upon release; and prostaglandins, which modulate smooth muscle activity.

2. How would you describe the mucosa of this organ, indicated with the red circle?
   The pseudostratified columnar epithelium is highly convoluted, forming a honeycomb appearance. The lumen may have villous projections protruding into it and may contain visible secretions. The epithelium lies on top of a lamina propria of connective tissue and thin vessels.

3. What is the structure in the blue circle? Given its proximity to the accessory organ, what do the cell bodies within it likely represent?
   This is an autonomic ganglion; presumably, a parasympathetic ganglion, given how close it is to its target organ. The preganglionic parasympathetic fibers would run in the pelvic splanchnic nerves (S2–4).

G. Prostate VS 1

Slide link

1. What is the luminal structure indicated by the black asterisk (*)? Describe its epithelial lining.
   This is the (prostatic) urethra, lined with urothelium (transitional epithelium).

2. Which prostate region is indicated by the red polygon? What disease is most strongly associated with this region of the prostate?
   This is the transitional zone, which is most associated with benign prostatic hyperplasia (BPH). Its proximity to the prostatic urethra explains the urinary symptoms (increased urinary frequency, need to arise at night to urinate, difficulty starting urinary stream, dribbling, etc.).

3. How might you describe the appearance of the prostatic glands, indicated with the blue rectangle? How many layers of glandular cells are there?
   The prostatic glands form irregular shapes, from simple loops to more complex serrated, angulated, and stellate shapes. Each gland is composed of two layers of cells: a basal layer that is proliferative, and a layer of glandular (secretory) cells. The absence of a basal cell layer suggests prostate adenocarcinoma.

4. Describe the tissue shown in the green circle. Given that there are essentially no glands in this region but almost entirely stroma, which region of the prostate is this?
   This is fibromuscular stroma. This is likely the anterior portion of the gland, which is composed almost entirely of stroma and few, if any, glands.

5. What are the structures indicated by the yellow arrows?
   These are corpora amylacea: lamellated (layered) concretions of inspissated secretions that may secondarily calcify. They are of no pathological significance.

H. Penis VS 1

Slide link

1. What erectile tissue structures are indicated by the black circles? Where do they lie within the penis?
   These are the corpora cavernosa, located on the dorsal side of the penis.

2. What are the structures indicated with the blue (dashed), black, and red arrows? From what parent structures do they arise?
   The blue dashed arrows indicate the deep dorsal vein of the penis, which drains into the internal pudendal vein.
   The red arrows indicate the dorsal arteries of the penis, which arise from the internal pudendal arteries.
   The black arrows indicate the dorsal nerves of the penis, which are branches of the pudendal nerves (S2–4).

3. What are the structures indicated with the green arrows? What tissue are they composed of?
   These are the tunicae albugineae of the corpora cavernosa and corpus spongiosum. They are composed of fibroelastic connective tissue.

4. What are the spaces indicated with the black asterisks (*)? How do they operate?
   These are vascular sinusoids of the corpora cavernosa, lined with thick fibromuscular walls. Blood flowing from the deep (cavernosal, central) arteries into the helicine arteries will flow out into these spaces lined with endothelium. This increases intracavernosal pressure, leading to compression of venous outflow. This establishes a steady state at an increased pressure, maintaining erection.

5. What erectile structure is indicated with the orange circle? What structure is at its center, marked with an orange asterisk (*)?
   This is the corpus spongiosum, with the penile (spongy) urethra at its center.

Answers to Study Questions:

These answers may not be all encompassing; rather they may be examples of the types of thought processes needed for application of the material.

1. Describe histologic features of the testes in post-pubertal individuals.

Seminiferous tubules are lined by a complex stratified epithelium (seminiferous epithelium) composed of Sertoli cells and germ cells at various stages of maturation. Germ cells, identified by location and nuclear morphology, include spermatogonia, spermatocytes, early/late spermatids, and spermatozoa (sperm). Primary spermatocytes are the most abundant germ cell observed within the seminiferous epithelium, because of the extended time required for completing prophase of meiosis I (about 22 days). Primary spermatocytes are large cells with compact chromosomes undergoing meiotic recombination. Spermatids are post-meiotic haploid germ cells. Early spermatids exhibit small round nuclei, whereas late spermatids near the lumen of the tubule exhibit condensed nuclear chromatin and flagella. The basement membrane of the seminiferous epithelium provides adhesion molecules for spermatogonia (mitotic germline stem cells) and supporting Sertoli cells (with distinctive almond-shaped nuclei). Myoid cells external to the basement membrane propel sperm and fluid from the testicular tubules to the excretory genital ducts. Leydig (interstitial) cells are testosterone-producing cells located in spaces between tubules.

2. Discuss the histology and significance of the blood–testis barrier.

Tight junctions between adjacent Sertoli cells divide the seminiferous epithelium into basal and luminal compartments. Testicular cells residing in the basal compartment of the epithelium include Sertoli cells and spermatogonia. Testicular cells located within the luminal compartment of the epithelium include meiotic and haploid germ cells (spermatocytes, spermatids, and spermatozoa). Tight junctions between adjacent Sertoli cells establish a blood–testis barrier that isolates developing germ cells within an immunologically privileged location. This blood barrier is important because developing germ cells express novel antigens, having passed through recombination. These novel antigens may be misinterpreted as "foreign" and trigger an autoimmune response (autoimmune orchitis).

3. Compare and contrast histologic features of the genital ducts involved in sperm transport and post-testicular sperm maturation.

Spermatozoa released from the seminiferous epithelium pass through intratesticular and extratesticular (excretory) genital ducts. Intratesticular ducts include the straight tubules and rete testis. Extratesticular ducts include the efferent ductules, epididymis, ductus (vas) deferens, ejaculatory ducts, and urethra. Spermatozoa continue to undergo post-testicular maturation during transport. Secretions of the paired seminal vesicles and the prostate gland

enhance sperm viability and motility following ejaculation. The rete testis is lined by simple cuboidal epithelium within the mediastinum testis. Efferent testicular ductules appear "scalloped" in cross section because of the presence of tall ciliated and smaller nonciliated epithelial cells. The epididymis (a single coiled tube) is characterized by the presence of stereocilia that are really very long, branched microvilli. The ductus deferens exhibits a pseudostratified epithelium and prominent muscularis. The portion of the ductus deferens that passes through the prostate, after having been joined by the secretions from the seminal vesicles, is referred to as the ejaculatory duct. The epithelium of the penile urethra transitions from pseudostratified to stratified squamous epithelium at the glans penis.

4. Compare and contrast histologic features of seminal vesicles and the prostate gland.

Paired seminal vesicles and the prostate are exocrine glands that provide fluid to semen. Both glands are lined by a pseudostratified columnar epithelium, composed of basal stem cells and secretory epithelial cells. The seminal vesicles contribute about 65–75% and the prostate gland contributes about 20–30% of the fluid of semen. Both glands feature a dense fibromuscular stroma. The sympathetic division of the autonomic nervous system innervates both glands. The seminal vesicle is a single, highly convoluted tube with extensive folding of the lining epithelium. By contrast, the prostate gland features branched tubuloalveolar glands that empty into sinuses alongside the prostatic urethra. Unlike the seminal vesicles, the prostate includes the prostatic urethra and the paired ejaculatory ducts. Mucosal glands surrounding the prostatic urethra in the "transitional zone" undergo benign prostatic hyperplasia as an individual ages, leading to physical compression of the prostatic urethra. Symptoms of nodular prostatic hyperplasia include urgency and frequency. Concretions of precipitated proteins may be visible within prostate glands. These corpora amylacea are of no clinical significance.